Tor Network

By Rice & Crackers

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Tor sends traffic through a chain of relays so that no single relay knows both who you are and where you are going. You are expected to explain that design and to run a non-exit relay the class can see carrying traffic. Topics to explore:
• How a circuit is built: guard, middle, and exit.
• Step by step setting up a Tor node in Linux.
• Directory authorities and the consensus.
• Bridges, pluggable transports, and onion services.
• A non-exit relay you run on a VPS or from your homelab.
• The relay listed as running on Tor Metrics, and traffic visible in Nyx.

Software: Tor Browser
Hardware: PC, Homelab Server or Raspberry Pi

What we expect to explore: Finding ways to search up websites that are being blocked by a firewall and searching the dark web

A short implementation plan:

1. Doing some research on Tor and how to set it up
2. Implementing our research into actually creating a Tor node using a homelab, RPi, or PC
3. Test it out by searching up blocked websites or searching up illegal websites like hiring a hitman

Tor Threat Model

Tor network:

- Designed to safeguard privacy and enable anonymous communication
- Designed to provide anonymity and privacy for its users.

Threat Model: Helps identify potential threats, assess risk, and design appropriate mitigation strategies to protect both users ad the network itself.

Together: The threat model is what the tor network is supposed to protect you from

Who is Tor used for:

- people who want to become anonymous while browsing the internet
- Ordinary users
- Journalists
- Activists
- People facing censorship

What Tor hides:
Tor hides the fact that your local isp cannot find what you are searching

- User's IP addres and browsing destination from certain observers
- Prevents websites and other services from learning your location, as they can use that data to build databases about your habits and interests

- Prevents people watching your local traffic, such as your ISP or someone with access to your home wifi or router, from learning information that you're fetching and where you're fetching it from
- Routes your connection through more than on Tor relay so no single tor node knows where the previous tor node/relay will know what you're doing

What Tor Does not hide
Tor does not hide your personal information if you explicitly put your information into a site. Your ISP is also able to see that you're using a Tor browser, though they won't be able to detect what sites you're using. In addition, while Tor is able to encrypt your data within its own network, there are some considerations to remember:

- HTTPS: the content of your connection is encrypted, so passwords, messages, and other data are protected
- HTTP: the content of your connections is not encrypted, so someone who is monitoring the connection will be able to read or modify transmitted information

How is a circuit built?

Every tor connection, which are circuits, are built from three relays / hops. First the client chooses 3 different nodes making sure that none of them repeat in the circuit.

The first hop / relay is the Guard/entry node: this node sees my real ip address but not the destination, and it receives my encrypted traffic and forwards it to the next hop.

The second hop/relay is the Middle node: and this node sees nothing, not my ip or the destination's, it sends the encrypted data between the guard and the exit.

The third hop/relay is the Exit node: this node sees the destination but not your real IP, it removes the final layer and connects to the website.

You → [Guard] → [Middle] → [Exit] → Website

Directory authorities and the consensus

The network relies on a small group of trusted nodes/servers in the Tor network known as Directory Authorities.
There are nine directory authorities, and their public keys are hardcoded into the Tor software.
They are operated by trusted volunteers, academic entities, and nonprofit organisations, and they monitor, test, and vote on the state of all active relays to publish and distribute a signed document (known as the consensus) containing information about all known Tor relays.

Bridges, pluggable transports, and onion services

- Tor bridges and pluggable transports and onion services help users bypass internet censorship and protect their privacy.
- Tor bridges are secret, unlisted Tor relay servers that do not appear in the public Tor directory.
- Standard Tor relay IP addresses are public, making it easy for governments or ISPs to block them. And since bridges are private, users can connect to the Tor network without their connection being flagged

- Countries that block Tor can block every public relay, so bridges give users a way to access Tor and browse safely, for example Tiago is hosting a bridge and he said he helped someone from Belarus.

- Pluggable transports hides or transforms the network traffic as it passes between you and the bridge
- Advanced censorship systems use Deep Packet Inspection (DPI) to recognize and block standard Tor signatures. Pluggable transports disguise Tor data so it looks like regular internet traffic
- Ex:
    - *obfs4*: makes traffic look like random data
    - *Snowflake*: Reroutes traffic through temporary volunteer proxies (like a browser extension) to look like standard WebRTC video or chat traffic

- Onion services are websites or applications accessible only within the Tor network using a .onion special-use domain
- Traffic never leaves the Tor network, so there is no exit relay
- They provide end-to-end anonymity and security for both the visitor and the service operator, hiding the physical location and IP address of the server hosting the site

Setting up a Tor node in Linux

We would choose either one of our raspberry PIs or a homelab pc to set up the Tor node and to leave it on, or if not we would use a vps, and then give it a public ip address. Then install Tor and Nyx, and then configure the node in the Tor config file, and give our node a nickname and a contact email and a port for it to listen on. Then set it as a non-exit relay so it doesn't send traffic out to websites. Then open the port in the firewall and start the Tor service.

Non-Exit Relay

- A non-exit relay is the server that helps carry Tor traffic without it being the last stop/final destination website.
- Tor sends each connection through three relays as mentioned before the guard, middle and exit
- Passes encrypted traffic to next relay, but never connects to website directly
- So it can't see what user is doing (IP never appears in traffic)
- You can run on a VPS which is a virtual private server provided by a hosting provider
    - Examples include: Linode, Hetzner, etc

Nyx

- Command line monitoring too used for Tor relays
- Ran on server
- Shows you live bandwidth, log messages, graphs, connections, etc
- Best and most efficient way to see if traffic passing through your relay

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